

Student Resource Guide

Generative AI for Beginners

Section 5: Audio and Video Generation | Lecture 5.2: Music and Audio Generation

Learning Objectives

- Distinguish between symbolic (MIDI-based) and audio-based approaches to music generation
- Explain how large-scale audio generation models (MusicGen, Suno, Udio) produce music from text prompts
- Identify the training data, architectural choices, and quality characteristics of major music generation systems
- Evaluate the commercial licensing implications of AI-generated music
- Describe applications of music and audio generation in commercial and creative production workflows

Introduction

AI-generated music has moved from a research curiosity to a commercially deployed content category in a remarkably short time. The launch of Suno and Udio in 2024 demonstrated that text-to-music generation had crossed a quality threshold sufficient for background music in commercial content—video production, advertising, games, and social media. Understanding how these systems work at a technical level equips practitioners to make informed decisions about when AI music generation is appropriate, what its quality characteristics and limitations are, and what intellectual property and licensing considerations apply.

Music generation presents unique technical challenges compared to image and speech generation. Music has complex hierarchical structure: individual notes combine into phrases; phrases combine into sections (verse, chorus, bridge); sections combine into complete songs. Emotional coherence must be maintained across this entire structure. Rhythm, harmony, and melody must coordinate across multiple simultaneous instruments. These requirements make music generation architecturally more complex than image synthesis.

The commercial implications are significant. AI music generation affects professional composers, music licensing markets, and content creators who previously required paid music libraries. Practitioners deploying AI music generation in commercial contexts must understand copyright status, licensing requirements, and the evolving legal landscape around training data and output ownership.

Key Concepts

Symbolic Music Generation (MIDI-Based)

Definition	Generating music as sequences of note events (pitch, duration, velocity, instrument) in MIDI format, rather than as audio waveforms. Models like Music Transformer and MuseNet generate MIDI sequences by learning patterns in large datasets of symbolic music, then predict subsequent events token by token.
Business Context	Symbolic generation produces editable, instrument-flexible output: a MIDI sequence can be rendered in any virtual instrument, transposed, tempo-adjusted, and edited in standard music production software (DAWs) without audio quality loss. This editability is highly valuable for professional music production workflows.
AI Practitioner relevance	Practitioners integrating AI music into production workflows prefer symbolic generation when the output needs to be edited, adapted, or combined with live instrument recordings. Audio-based generation is appropriate when the final audio output is the deliverable and editing is not required.
Real-World Example	A game studio uses MuseNet to generate procedural background music variations in MIDI format. The MIDI output is imported into their game engine, where the game's audio middleware renders it through different virtual instrument sets depending on the in-game environment—the same AI-generated melody sounds different in a dungeon (dark orchestral) vs. a village (folk instruments).

Audio-Based Music Generation

Definition	Generating music directly as audio waveforms or codec tokens, encompassing both instrumental content and vocals. Models like MusicGen (Meta), Suno, and Udio use variants of transformer architectures and audio codecs (EnCodec, SoundStream) to generate multi-second musical sequences from text prompts.
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Business Context	Audio-based generation delivers immediately playable, production-quality audio without requiring music production software or expertise. For content creators, marketers, and developers who need background music quickly and without musical training, audio generation tools provide immediate value.
AI Practitioner relevance	Practitioners evaluating audio music generation tools should assess: maximum clip length, continuation capability (extending generated clips), stem separation (can individual instruments be isolated?), vocal quality (if vocals are included), and commercial licensing terms.
Real-World Example	Suno v3 (2024) can generate a complete 2-minute song with sung vocals, backing instrumentation, and structure (verse-chorus-verse) from a text prompt like 'upbeat indie pop song about summer adventure, female vocals'. The output is production-quality audio ready for use as video background music or social media content.
Key Risks	AI music generators trained on copyrighted music may produce outputs that resemble specific copyrighted works. Several major music labels have filed copyright infringement suits against AI music generation companies (2024). The legal status of AI-generated music training and output is actively litigated.

Audio Codec Tokenisation

Definition	A technique that represents audio as discrete tokens using a neural audio codec (EnCodec, SoundStream). The codec compresses audio into sequences of integer tokens; the generative model predicts token sequences like a language model predicts word sequences; the codec decoder converts token sequences back to audio.
Business Context	Audio codec tokenisation allows music generation to use transformer architectures directly—the same class of models that power large language models—applied to audio token sequences.

	This enables leveraging the entire LLM research ecosystem (scaling laws, training techniques, inference optimisation) for audio generation.
AI Practitioner relevance	Understanding codec tokenisation explains why music generation models have improved so rapidly: they benefit directly from advances in transformer architecture and training at scale. The same efficiency improvements that make LLMs faster and cheaper also apply to codec-based audio generators.
Real-World Example	Meta's MusicGen uses EnCodec to represent audio as four parallel token streams at different temporal resolutions. A transformer generates these token streams autoregressively; EnCodec decodes them to stereo audio. This architecture enables generating up to 30 seconds of high-quality music from text prompts.

Commercial Licensing and IP Considerations

Definition	The legal framework governing the use of AI-generated music in commercial contexts, including questions about training data copyright, ownership of AI-generated outputs, and the terms of use imposed by specific music generation platforms.
Business Context	Using AI music without understanding licensing exposes organisations to copyright infringement liability. Terms vary dramatically: some platforms grant full commercial use rights (e.g., Suno's paid plans); others restrict use to personal projects; some require attribution. Additionally, the training data copyright issue remains unresolved in multiple jurisdictions.
AI Practitioner relevance	Practitioners procuring AI music for commercial use should: verify the platform's commercial licensing terms, understand whether the platform has licensed its training data, assess the risk level appropriate for the use case (low-stakes internal vs. high-profile advertising), and consider whether a traditional royalty-free music library provides better legal

	certainty.
Real-World Example	A marketing team used AI-generated music in a TV commercial without verifying the platform's commercial licensing terms. The platform's terms of service restricted outputs to personal use only. The team had to remove the ad and replace the music—incurring both production costs and legal review fees.

Real-World Applications

AI music and audio generation is being deployed across content creation, game development, and production industries at increasing scale.

Video Content Background Music: Content creators, video producers, and social media marketers use AI music generation to produce custom background music for each video—matching the specific emotional tone, pacing, and style without paying per-track licensing fees or commissioning composers. The ability to specify exact mood and style through text prompts is particularly valuable.

Adaptive Game Music: Game developers use music generation to create adaptive soundtracks that respond dynamically to player actions and emotional states. AI-generated music that shifts mood based on in-game tension, player progress, or time of day creates more immersive experiences than looped pre-composed tracks.

Podcast and Broadcast Production: Podcast producers use AI-generated music for intros, outros, and segment transitions that are unique to their brand without paying composer fees. The ability to specify instrumentation, tempo, and genre through text enables non-musician producers to control music aesthetics precisely.

Sound Effects and Foley Generation: AI audio generation extends beyond music to sound effects: generating specific ambient sounds, UI audio feedback, and foley effects from text descriptions. Tools like AudioCraft (Meta) and ElevenLabs Sound Effects generate custom audio that would otherwise require studio recording sessions.

Best Practices

Verify commercial licensing terms before production deployment. Never assume AI music is freely usable commercially. Check whether the specific plan/tier you are using grants commercial rights, and whether these rights apply to your type of use (broadcast, product, advertising).

Generate longer clips and select the best section. Most AI music generators perform unevenly across the full duration of a clip. Generate 2x the required length and select the strongest portion rather than using the full output.

Use audio generation for prototyping, symbolic generation for production. AI-generated audio is excellent for rapid concept approval. When content moves to production, consider whether a musician or producer should use the AI output as inspiration rather than final product.

Test audience response before scaling AI music use. AI music often lacks the emotional depth of human-composed music. Test listener response, particularly in high-emotion contexts (advertising, memorial content, brand storytelling), before committing to AI music at scale.

Common Mistakes to Avoid

Mistake: Assuming AI-generated music has no copyright issues because 'AI created it'.

Correct approach: AI music generation platforms impose their own licensing terms on outputs. Additionally, the underlying training data copyright status is actively contested. Review specific platform terms and obtain legal advice for high-stakes commercial use.

Mistake: Using the first generated clip without evaluation for quality consistency.

Correct approach: Music generators produce variable-quality outputs. Generate 3–5 variations of any music prompt and select the best. Evaluate for musical coherence, appropriate energy level for the use case, and absence of artifacts or abrupt transitions.

Mistake: Expecting AI music generation to understand precise musical structure specifications.

Correct approach: Current models respond well to genre, mood, and instrumentation prompts but have limited ability to follow precise structural specifications (e.g., 'start with 8-bar intro, then 16-bar verse'). Use symbolic generation and DAW editing for precise structural control.

Practitioner Tool: AI Music Generation Tool Comparison

Type: Comparison Table

Reference this table when selecting an AI music generation tool for a specific production context.

Tool	Output Type	Max Length	Vocals?	Commercial Rights?	Best Use Case
Suno v3/v4	Audio	2-4 min	Yes	Paid plans: yes	Complete songs,

	(mp3)				content creation
Udio	Audio (mp3)	~2 min	Yes	Check plan terms	Experimental, artistic
MusicGen (Meta)	Audio	30 sec	No	CC-BY-NC (research)	Research, background music
MuseNet (OpenAI)	MIDI	Variable	No	Check ToS	Orchestral, DAW integration
AudioCraft (Meta)	Audio/SFX	~30 sec	No	CC-BY-NC	Sound effects, audio research
Stable Audio	Audio	~3 min	No	Paid: commercial	Instrumental background music

Note: Verify current commercial terms directly with each vendor before production use—licensing terms change frequently as the legal landscape evolves.

Knowledge Check

Question 1: A game studio needs AI-generated music that can be transposed, edited in their DAW, and rendered through different virtual instruments for different game environments. Which generation approach is most appropriate?

- A. Audio-based generation using Suno for direct playback
- **B. Symbolic (MIDI-based) generation for full editability and instrument flexibility**
- C. Voice cloning technology applied to a composer's hummed melodies
- D. Codec tokenisation with EnCodec for maximum audio quality

Correct Answer: B. Symbolic (MIDI-based) generation for full editability and instrument flexibility

Explanation: MIDI output is editable in standard DAWs, transposable, and can be rendered through any virtual instrument. Audio-based generation produces fixed audio waveforms that cannot be meaningfully edited for instrument changes or transposition.

Question 2: A marketing team plans to use AI-generated music from a free-tier account in a national television commercial. What is the most significant risk they should evaluate first?

- A. The audio quality may be insufficient for television broadcast
- **B. Free-tier accounts on most AI music platforms restrict commercial use—they likely lack the required commercial rights**
- C. AI music cannot exceed 30 seconds in length
- D. The music will be identifiable as AI-generated by viewers

Correct Answer: B. Free-tier accounts on most AI music platforms restrict commercial use—they likely lack the required commercial rights

Explanation: Most AI music platforms restrict commercial rights to paid plans. Using free-tier outputs in a national television commercial likely violates the platform's terms of service, exposing the company to licence violation liability.

Reflection Questions

Take time to consider the following questions before the next session:

1. AI music generation raises questions about creative authorship—if an AI composes music, who is the creator? How should attribution work in commercial contexts?
2. Professional composers and musicians face economic disruption from AI music generation. How should the industry respond—through licensing frameworks, royalties, or other mechanisms?
3. The ability to generate music in any artist's 'style' (without using their recordings) raises questions about the limits of copyright. Where should style protection end?
4. Adaptive AI music in games and interactive media could respond to player emotion and context in real time. What new design possibilities does this open, and what are the technical barriers?
5. As AI music quality improves, what will distinguish 'valuable' music from 'generic' music? How does this change what human composers should focus on?

Key Takeaways

- Music generation uses two fundamentally different approaches: symbolic (MIDI-based, editable) and audio-based (directly playable, immediate).
- Audio codec tokenisation (EnCodec) allows audio generation to leverage transformer architectures from the LLM ecosystem, explaining rapid quality improvements.
- Major tools (Suno, Udio, MusicGen) generate production-quality audio from text prompts describing genre, mood, instrumentation, and vocals.
- Commercial licensing terms vary significantly across platforms and tier levels—verify before any production deployment.
- AI music works best for background music, content creation, and prototyping; human composers still provide superior emotional depth for high-stakes creative contexts.
- The legal landscape around AI music copyright (training data, output ownership) is actively contested and should be monitored quarterly.

Next Actions

Complete at least two of the following before the next lecture:

6. Generate the same music prompt ('upbeat corporate background music, no vocals, 60 BPM, piano and strings') on Suno and a royalty-free music library (e.g., Epidemic Sound)—compare quality and cost.
7. Review Suno's current terms of service for commercial use rights and document the specific restrictions and requirements.
8. Listen to a MuseNet-generated orchestral piece (available on OpenAI's website) and evaluate it against a human-composed piece in the same style for structural coherence and emotional depth.
9. Research one pending copyright lawsuit involving AI music generation and summarise the core legal claim—understanding the legal landscape helps anticipate future compliance requirements.